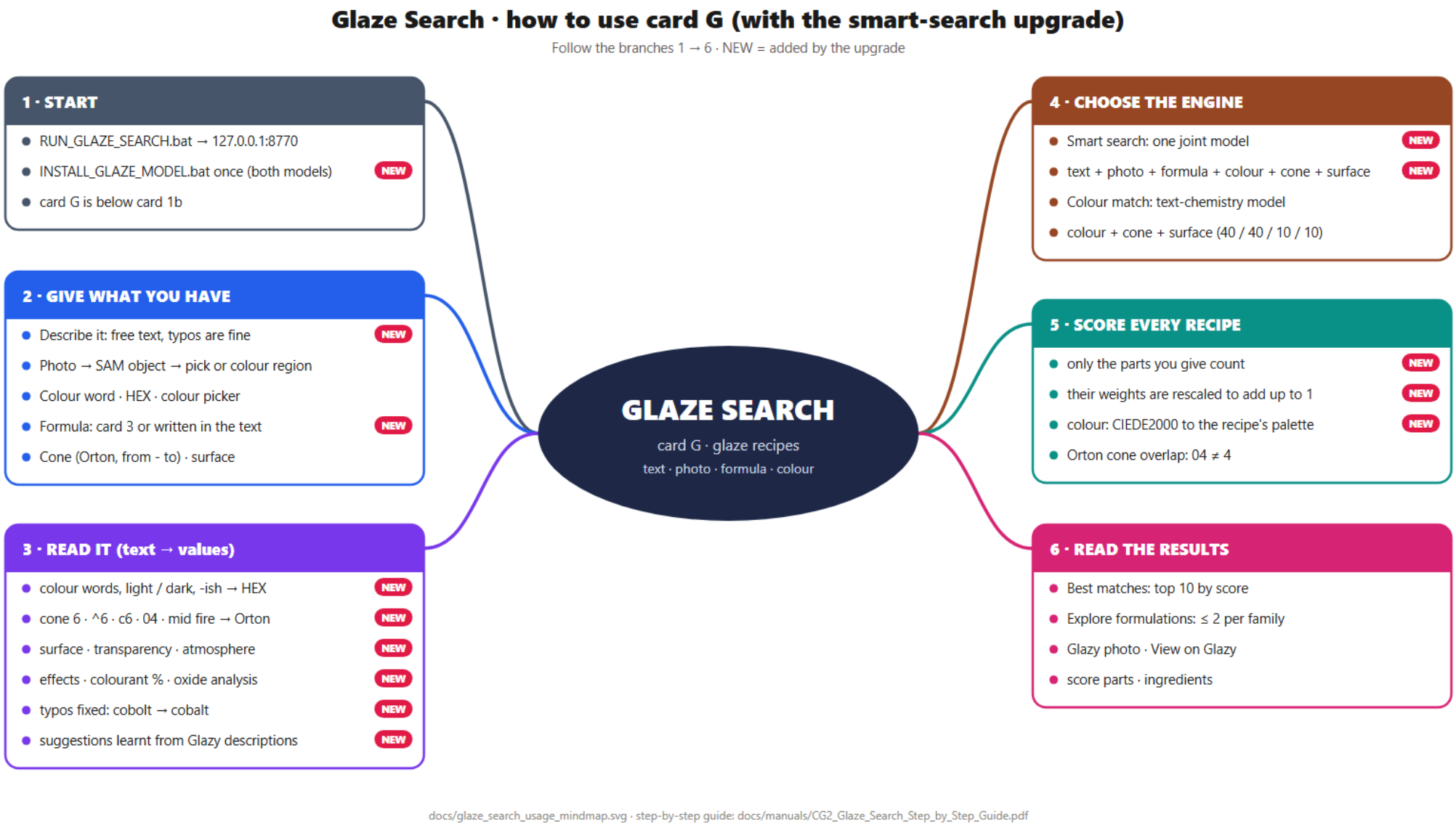
Glaze Search · how a search works (card G)

Usage mind map with the smart-search upgrade · NEW = added by the upgrade



docs/glaze_search_usage_mindmap.svg · step-by-step guide: docs/manuals/CG2_Glaze_Search_Step_by_Step_Guide.pdf

The process step by step · model and parameters of each step

Steps 1–5 run when you use a photo (card 1); 6–10 run for every card G search. Frozen = pretrained, not changed; trained = learnt on the 7k recipes.

1 Photo in

card 1 · app/server.py

- upload, paste, drop or link
- EXIF rotation, longest side ≤ 2,048 px
- saved under work\queries\<id>

2 Find objects

Grounding DINO base

IDEA-Research/grounding-dino-base

- prompt: ceramic object · pottery piece · ceramic test tile · … · glazed pottery
- box threshold 0.24 · text threshold 0.25
- NMS IoU 0.60 · hands excluded (mask IoU > 0.30)

3 Outline

SAM 2.1 Hiera-Large

facebook/sam2.1-hiera-large

- box prompt → exact mask per object
- label map: hover / click an object

4 Ceramic?

OpenCLIP ViT-B/32

laion2b_s34b_b79k

- p(ceramic) ≥ 0.35 → selected by default
- else the first object

5 Colour

pick · regions · palette

- pick: 13 × 13 px inside the mask, drop brightest / darkest 15 %, median
- regions: SLIC 220 superpixels (compactness 10) ≤ 320 px, merge ΔE76 < 11, min 1.2 %, counted from 4 %, median CIELAB
- names: nearest of 98 reference shades (CIEDE2000)
- recipe palette: ColorThief top 3

6 Read the text

rules + fuzzy matching · tag model

- typos: difflib ratio ≥ 0.80 (≥ 0.85 for ≤ 4 letters)
- 38 Orton cones (04 ≠ 4) · 9 surfaces · 4 transparencies · 7 atmospheres · 10 effects · 22 colourants · 26 oxides
- low / mid / high fire → 06–04 / 5–6 / 9–10
- suggestions: TF-IDF (word 1–2 + char 3–5 grams) → logistic regression C = 4, balanced; shown at p ≥ 0.45

7 Encode

frozen pretrained encoders

- text: all-MiniLM-L6-v2, mean pooled, 384-d, ≤ 128 tokens
- photo: DINOv2 ViT-L/14, 1024-d, area-weighted mean of the objects
- formula: 26 oxides → fractions → √ (26-d)

8 Joint model

3 trained heads → one 256-d space

- Linear → GELU → Dropout 0.1 → Linear (hidden 512) → L2 norm
- InfoNCE τ 0.07: text–photo, text–formula, photo–formula
- AdamW lr 1e-3 · wd 1e-4 · batch 256 · early stop (patience 12): best epoch 58

9 Score

every recipe, parts you give

- text .35 · photo .35 · formula .25 · colour .30 · cone .10 · surface .10 · transp. .05 · atmos. .05 → rescaled to 1
- similarity (cos + 1) / 2 · colour exp(−ΔE00 / 20) to the closest palette colour ≥ 10 %
- colour match: .40 latent + .40 colour + .10 surface + .10 cone

10 Results

two lists from one ranking

- Best matches: top 10
- Explore: ≤ 2 per family (K-Means k = 10 on the joint space)
- Glazy photo m_ → l_ · View on Glazy · ingredients

segmentation (pretrained) · colour measurement · text understanding · frozen encoders · trained joint model · scoring

Smart search model · training and held-out test results

model 78afd7dfbe2c · trained 2026-11-04 00:02:20 · 7,063 recipes (7,052 with photo objects, 7,049 with a formula, 4,766 with a description)

Training

data split	4,945 train · 1,059 validation · 1,059 test recipes, by composition group (near-identical recipes stay in one part)
text the model learns from	per step: tag text 25 % (colour word + cleaned tags, Orton cones kept), the recipe's own Glazy description 20 %, a noisy sentence 55 % (8 per recipe: typos, shorthand, filler words, any order)
loss	symmetric InfoNCE, temperature 0.07, for text–photo, text–formula, photo–formula; recipes of the same composition group are not negatives
optimiser	AdamW, learning rate 1e-3, weight decay 1e-4, batch 256, up to 150 epochs, early stop after 12 epochs without gain on validation (best epoch 58 of 70)
families (Explore)	K-Means on the fused 256-d embeddings, k from 8 / 10 / 12 / 15 / 20 by silhouette → k = 10 (silhouette 0.10). HDBSCAN for comparison: 2 clusters, 35 % noise - no clear density groups, so it is not used
text suggestions	surface 63 % (always "glossy" 53 %) · transparency 68 % (48 %) · atmosphere 93 % (88 %) · colour family 37 % (37 %, not used)

Reading noisy text (1,059 sentences of test recipes, 15 % typos)

	colour (ΔE00 ≤ 20)	cone	surface	atmosphere
right when read	93.3 %	99.2 %	89.1 %	100.0 %

Finding the right recipe (test split, gallery of 1,059 recipes, random top 10 ≈ 0.9 %)

query → recipe	n	top 1	top 5	top 10	median rank
noisy sentence → recipe, all parts (text + photo + formula)	1,059	31.7 %	61.9 %	**74.2 %**	3
noisy sentence → recipe text, pretrained MiniLM only (no training)	1,059	24.6 %	50.1 %	**62.0 %**	5
photo → recipe (text + formula side)	1,058	14.3 %	34.9 %	**45.3 %**	14
tag text → recipe (photo + formula side)	1,059	9.9 %	28.6 %	**40.2 %**	16
real Glazy description → recipe	713	7.7 %	23.8 %	**33.4 %**	31
formula → recipe (text + photo side)	1,058	6.6 %	23.7 %	**33.1 %**	26
noisy sentence → recipe (photo + formula side only)	1,059	7.2 %	18.1 %	**28.2 %**	33

Full smart search on noisy sentences (300 test sentences)

scored with	top 10 matching colour + surface + cone
text model only	54.4 %
read values only (colour, cone, surface …)	55.3 %
both = smart search	**73.1 %**

All numbers are measured on recipes the model never saw in training or model selection (scripts/train_glaze_joint.py writes them to models\glaze_joint\meta.json). Noisy sentences are generated from the recipes' tags; real user text can differ.

The whole system · three flows

Flow 1 builds the database once · Flow 2 searches objects by photo · Flow 3 is card G

